Nonuniqueness of induced-norm-minimizing right inverses

Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.

Abstract

The same 2×3 full-row-rank matrix has multiple right inverses minimizing its induced $\ell^p \to \ell^2$ norm for every $2 < p < \infty$, over both fields. The complete minimizer set is a disk over the complex field and an interval over the reals. A complex higher-dimensional construction gives a ball of minimizers. The smallest matrix is attributed in the recovered notes to the motivating paper's spectral-norm example; its construction is not claimed as new.

1 Pointwise orthogonality

For full-row-rank $A \in \mathbb{C}^{m\times n}$, $m < n$, consider

$$\min_{AY=I_m} \|Y\|_{p\to 2}\,, \qquad \|Y\|_{p\to 2} = \sup_{z\neq 0} \|Yz\|_2 \,/\, \|z\|_p\,.$$

If $B = A^\dagger$, every right inverse is $Y = B + N$ with $AN = 0$. Since $B = A^*(AA^*)^{-1}$, $B^*N = 0$, and therefore

$$\|Yz\|_2^2 = \|Bz\|_2^2 + \|Nz\|_2^2\,.$$

Thus B minimizes the objective for every p. This is not the different product objective $\|YA\|_{p\to 2}$.

2 Smallest counterexample

Theorem 1. *Let*

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}, \quad B = \tfrac{1}{3}\begin{pmatrix} 1 & 1 \\ 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Then $B = A^\dagger$, $AB = AX = I_2$, $B \neq X$, and for $2 \le p \le \infty$,

$$\boxed{\|B\|_{p\to 2} = \|X\|_{p\to 2} = \min_{AY=I_2} \|Y\|_{p\to 2} = 2^{1/2-1/p}.}$$

Proof. Direct multiplication gives $B^*B = \frac{1}{3}\left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right) \preceq I_2$ and $X^*X = I_2$. The inequality $\|z\|_2 \le 2^{1/2-1/p}\|z\|_p$ gives both upper bounds. At $z = (1,-1)^T$ both matrices preserve Euclidean norm, so the bounds are attained. Pointwise orthogonality proves global optimality. $\square$

When $m = 1$, the right inverse is a single column, and its induced norm is its Euclidean norm for all p; orthogonality makes the minimum unique. Hence dimensions 2×3 are smallest.

3 Complete minimizer set

Every right inverse is $B+v(a,b)$, where $v=(-1,1,1)^T$. At norming vector $(1,-1)^T$, orthogonality forces $a=b=t$ for any minimizer. The Gram matrix of $Y=B+tv(1,1)$ has eigenvalue one on $(1,-1)^T$ and eigenvalue $1/3+6|t|^2$ on $(1,1)^T$. Both vectors have equal-modulus coordinates. Testing the latter forces $|t|\le 1/3$. Conversely this condition gives $Y^*Y \preceq I_2$ and the same attained bound. Thus for every $2\le p\le\infty$, all minimizers are

$$\boxed{B+t(-1,1,1)^T(1,1), \qquad |t|\le \tfrac{1}{3}.}$$

The real parameter interval is $[-1/3,1/3]$, and X corresponds to $t=1/3$.

4 Every complex dimension

For $2\le m<n$, take $A=(\mathbf{1}\ \ I_m\ \ 0_{m\times(n-m-1)})$, $B=A^\dagger$. Then

$$B^*B = I_m - \frac{\mathbf{1}\mathbf{1}^*}{m+1} \preceq I_m.$$

The $m-1$ nonconstant Fourier vectors $(1,\omega^j,\ldots,\omega^{(m-1)j})^T$, $1\le j<m$, $\omega=e^{2\pi i/m}$, have flat coordinate moduli and span $\mathbf{1}^\perp$. The matrix B preserves their Euclidean norms. Together with $\|z\|_2 \le m^{1/2-1/p}\,\|z\|_p$, this proves optimum $m^{1/2-1/p}$ and makes those vectors norming vectors.

For an optimal $Y=B+N$, orthogonality forces N to vanish on every Fourier vector, hence on $\mathbf{1}^\perp$. Therefore $N=u\mathbf{1}^*$ with $u\in\ker A$. Its Gram matrix is

$$Y^*Y = I_m + \left(\|u\|_2^2 - \frac{1}{m+1}\right)\mathbf{1}\mathbf{1}^*.$$

Testing the flat vector $\mathbf{1}$ shows that $\|u\|_2 \le (m+1)^{-1/2}$ is necessary, and $Y^*Y\preceq I_m$ shows sufficiency. Thus the complete complex minimizer set is

$$\boxed{B+u\mathbf{1}^*, \quad u\in\ker A, \quad \|u\|_2\le (m+1)^{-1/2}.}$$

This higher-dimensional statement uses complex Fourier vectors. The smallest counterexample is valid over either field.

5 Attribution and verification

The rational verifier checks the right-inverse, Gram, and kernel identities. Continuous parameter ranges and complete minimizer sets are analytic claims. The recovered notes attribute the smallest A,B to Example 4.1 of Dokmanić and Gribonval for the spectral norm; that construction is not claimed as new, and the attribution should be checked before publication.

Recovered references: OpenProblemsInNLA, provisionally IE-23; I. Dokmanić and R. Gribonval, *Beyond Moore–Penrose, Part I: Generalized Inverses that Minimize Matrix Norms*, arXiv:1706.08349v2 (2017), Example 4.1, Corollary 4.2(3), Remark 4.1.[1] The live repository statement was not independently retrieved during reconstruction.

[1] `https://arxiv.org/abs/1706.08349`. References retained as source leads; priority and exact source scope require checking.